SERGIO TASCON MORALES

Ph.D. in Biomedical Engineering

INFO

E-mail
s.tasconmorales@gmail.com

Website/GitHub
sergiotasconmorales.github.io
/sergiotasconmorales

Nationality
Colombian

Marital status
Single

Residence permit
Type B
You are allowed to hire me

KEY COMPUTER SKILLS

Programming Languages
Python
C/C++
Matlab

Libraries / Frameworks
PyTorch
PyTorch Lightning
Tensorflow + Keras
HuggingFace
LangChain
OpenCV
NLTK
NumPy
SpaCy
Scikit Learn
Pandas

Tools / Platforms
Bazel
Git
AWS
SLURM
VSCode
ITK-Snap
LaTex

PROFILE

I am a highly skilled and ambitious professional seeking new opportunities in the field of Machine Learning, especially with focus on Computer Vision and/or NLP. Known for my structured approach and meticulous attention to detail, I consistently deliver high-quality results within specified timelines. My strong teamwork abilities enable me to thrive in collaborative environments, where I value and integrate diverse perspectives, personalities, and cultural backgrounds to drive innovative solutions.

WORK EXPERIENCE

11.2024 - 04.2025
Zurich, CH

Postdoctoral Researcher / Senior ML Engineer
ETH Zurich / Scanvio Medical

- Contributed to the development of a deep learning-based diagnostic platform for endometriosis, enhancing the analysis of ultrasound data for improved accuracy.
- Implemented and optimized features in the codebase, ensuring seamless functionality and alignment with project goals.

07.2024 - 10.2024
Bern, CH

Research Assistant (part-time 60%)
University of Bern

- Conducted advanced research in Medical Visual Question Answering, leveraging foundation models and Retrieval-Augmented Generation (RAG) to enhance diagnostic and analytical capabilities.
- Supervision of a master student in a project about Visual Question Answering for ECG signals.

02.2020 - 10.2020
Heidelberg, DE

Research Intern
Mediri GmbH

- Developed my master thesis on segmentation of multiple sclerosis lesions from longitudinal data using a combination of CNNs and RNNs.

03.2016 - 07.2016
Maulburg, DE

Research Intern
Endress & Hauser

- Evaluated Python as an alternative for future use in the company and conducted analysis of models for pressure sensor linearization..

EDUCATION

11.2020 - 05.2024
Bern, CH

Ph.D. in Biomedical Engineering
University of Bern

- Developed solutions in the topic of Visual Question Answering (VQA), a multi-modal task involving images, questions and answers.
- Thesis title: Spatial Awareness and Logic for Robust Visual Question Answering.

09.2018 - 09.2020
Le Creusot, FR
Cassino, IT
Girona, ES
Heidelberg, DE

Erasmus+ Joint Master in Medical Imaging and Applications (MAIA)
University of Burgundy, University of Cassino, University of Girona

- Major projects include:
  - Left ventricle segmentation using K-Means and Active Contours
  - Classification of Chest X-Ray images using Deep Learning

COMPETENCES

AI Skills

General machine learning
Convolutional neural networks (CNN)
Transformers
Vision transformers (ViT)
Large language models (LLM)
Retrieval-augmented Generation (RAG)
Multimodal LLMs
Parameter efficient finetuning (PEFT)
Recurrent neural networks (RNN)
Dataset creation
Automatic report generation
Computer aided diagnosis (CAD)
Image segmentation
Image registration

General

Time management
Problem solving
Teamwork
Adaptability
Creativity
Independent work
Critical thinking
Attention to detail

LANGUAGES

Spanish	*Native*
English	*Advanced (C1)*
German	*Advanced (C1)*
Swiss German	*Intermediate (B1)*
Italian	*Intermediate (B1)*
Turkish	*Basic (A1)*
Russian	*Basic (A1)*

REFERENCES

Dr. Fabian Laumer
CTO, Scanvio Medical AG
Zurich, Switzerland
fabian.laumer@scanvio.com

Prof. Dr. Raphael Sznitman
ARTORG Center for Biomedical Research Director
University of Bern
Bern, Switzerland
raphael.sznitman@unibe.ch

Dr. Johannes Gregori
Research Director
Mediri GmbH
Heidelberg, Germany
j.gregori@mediri.com

HOBBIES

Violin
Hiking
Drawing
Reading
Football

- Segmentation and classification of breast mass lesions using grayscale morphological operations and decision trees
- Classification of skin lesions using classical machine learning and deep learning
- 3D Harris corner detector

> Thesis title: Multiple Sclerosis Lesion Segmentation Using Longitudinal Normalization and Convolutional Recurrent Neural Networks.

10.2015 - 03.2016
Ilmenau, DE

Exchange Semester
Ilmenau University of Technology

> Exchange semester during Young Engineers program of the German Academic Exchange Service (DAAD).

02.2011 - 12.2017
Cali, CO

Electronic Engineering
Universidad del Valle

> Main focus on machine learning and computer vision.

HONORS & AWARDS

2020
Le Creusot, FR
Best master thesis award
University of Burgundy, University of Cassino, University of Girona

2020
Le Creusot, FR
Best student award
University of Burgundy, University of Cassino, University of Girona

2018
Girona, ES
Erasmus+ grant
European Union

2017
Cali, CO
Bachelor final project with distinction
Universidad del Valle

2015
Cali, CO
Scholarship Young Engineers
German Academic Exchange Service (DAAD)

2009
Ginebra, CO
Best middle school graduate
Institución Educativa Inmaculada Concepción

PUBLICATIONS

2024
Targeted Visual Prompting for Medical Visual Question Answering
AMAI Workshop at the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: P. Márquez-Neila, R. Sznitman.

2023
Localized Questions in Medical Visual Question Answering
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: P. Márquez-Neila, R. Sznitman.

2023
Logical Implications for Visual Question Answering Consistency
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). Co-authors: P. Márquez-Neila, R. Sznitman.

2022
Consistency-Preserving Visual Question Answering in Medical Imaging
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: P. Márquez-Neila, R. Sznitman.

2020
Multiple Sclerosis Lesion Segmentation Using Longitudinal Normalization and Convolutional Recurrent Neural Networks
MLCN Workshop at the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Co-authors: S. Hoffman, M. Treiber, D. Mensing, A. Oliver, M. Günther, J. Gregori.